

Research paper

Feasibility design, comparative evaluation, and energy consumption prediction of standalone hybrid energy system for rural electrification in Ghana using machine learning models

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ABSTRACT

This study investigated the feasibility and sustainability of standalone hybrid energy systems for rural electrification in Ghana. The problem addressed was the lack of electricity access in rural areas of Ghana, despite progress in increasing access rates in urban areas. The importance of this study lies in identifying a reliable, affordable, and environmentally sustainable solution to bridge the electrification gap. This study employed a comprehensive analysis of different system configurations, including Solar Photovoltaic (SPV), Diesel Generators (DG), and Battery Storage (BS), using the Hybrid Optimization of Multiple Electric Resources (HOMER) software. The integration of machine learning models into the design process was also explored to enhance the accuracy of energy consumption predictions. The economic, technical, and environmental aspects of each alternative are evaluated. The significant results obtained from the analysis showed that the optimal design for the standalone hybrid energy system consisted of a 112 kW SPV array, 80 kW DG, 342 kWh BS, and a 54-kW converter. The Levelized Cost of Energy (LCOE) obtained from this configuration was \$0.0263/kWh, with a renewable fraction of 91.5 %. The Net Present Cost (NPC) of the system was also the lowest among the alternatives, amounting to \$244,937. Further investigation of the Electric Distance Limit (EDL) of the optimal solution is 7.5 km which depicts that, although grid-connected electricity may seem more affordable, the proposed standalone hybrid energy system is a more viable option for remote communities that are far from the grid and makes it a suitable solution for addressing the electricity access gap in Ghana's rural areas.

1. Introduction

The United Nations has set an ambitious target to achieve 100 % access to electricity by 2030 [1]. Between 2010 and 2020, the global electricity access rate increased from 83 to 91 %. Simultaneously, the population lacking electricity access has decreased significantly, declining from a staggering 1.2 billion to 733 million [2]. In Africa, the

annual growth rate of the population with electricity access nearly tripled, experiencing a remarkable surge, escalating from 8 million between 2000 and 2013, to a staggering 24 million between 2014 and 2019. As a result, the number of people lacking electricity access, which reached a peak of 613 million in 2013, steadily declined to 572 million in 2019 [3]. Africa's population of 572 million represents a significant portion of the 733 million people without electricity worldwide.

Sub-Saharan Africa has the lowest electricity access rate worldwide,

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Nomenclature			
AA	Alternative A	HRES	Hybrid Renewable Energy System
AB	Alternative B	kW	Kilowatt
AC	Alternative C	MW	Megawatt
AD	Alternative D	LCOE	Levelized Cost of Energy
BS	Battery Storage	NPC	Net Present Cost
CC	Capital Cost	O&M	Operation and Maintenance
DG	Diesel Generator	PURC	Public Utility Regulatory Commission
EC	Energy Commission	PV	Photovoltaic
GDP	Gross Domestic Product	RE	Renewable Energy
GHG	Greenhouse gas	SDG	Sustainable Development Goals
HES	Hybrid Energy System	SPV	Solar Photovoltaic
HOMER	Hybrid Optimization for Multiple Electric Renewables	SSA	Sub-Saharan Africa
		UN	United Nations

with just over two-fifths of the population accessing it. According to World Bank data, the electricity access rate in Africa as of 2020 according to World Bank data stands at 48.2 %, which, given the level of income and electric grid footprint, is far less than it could be. Lack of access to electricity significantly hinders public service delivery, the use of new technologies, and quality of life [4,5]. In this context, Ghana stands out as an SSA country with a relatively higher electricity access rate of 86 % [6]. However, there is a significant disparity between urban and rural electrification rates in Ghana, with urban areas enjoying an access rate of 94.7 % while rural areas lag at 74 % [4,5].

Ghana's power generation mix comprises a combination of hydro-electricity, solar energy, and thermal energy from sources such as natural gas, crude oil, and diesel, as well as imports from neighbouring countries such as the Ivory Coast [6,7]. Despite the availability of diverse energy sources, the challenge lies in extending the national grid to remote and isolated areas that are far from the existing grid infrastructure. The financial feasibility of such grid extensions is a significant constraint under these circumstances.

Recognizing the need to address the energy access gap and align with international commitments, Ghana has embraced the United Nations SDG-7, which strives to achieve the widespread availability of reasonably priced, dependable, sustainable, and contemporary energy for every individual [8]. Additionally, Ghana has committed to its National Determined Contribution (NDC) goals, including a target of adding 10 % RE into the energy mix of the country by 2030 [8–11]. Achieving these ambitious targets necessitates innovative solutions, particularly in rural settings, wherein extending the national grid is not economically viable.

Mini-grids have emerged as a promising solution for electrifying these rural communities and accelerating progress towards national energy goals. Ghana possesses abundant renewable energy resources [12], including solar, wind, biomass, and hydropower, which surpass the country's energy demand. Leveraging these renewable energy sources through mini-grid systems can provide sustainable and efficient means of electrification, enabling rural communities to access reliable and affordable electricity.

Optimization of hybrid energy systems would be needed to realize sustainable rural electrification in developing countries like Ghana. Accurate prediction of the energy use is a fundamental but very crucial step in the design process of such systems. Traditionally, energy demand forecasting is done using statistical techniques, but recent developments in machine learning make it a potential third option to further improve the accuracy of predictions. This work integrates machine learning models into the design and optimization process for hybrid energy systems to improve the reliability and efficiency of rural electrification projects.

Several studies have been conducted on the ideal design of hybrid microgrid energy systems and their potential. This is generally achieved using metaheuristic optimization algorithms, which play a crucial

technical role in the optimal configuration. A study by [13] evaluated the viability of installing hybrid renewable energy systems (HRES) in the Philippines. The study found that HRES are both economically and environmentally advantageous compared to diesel-only systems, especially on larger islands where wind turbines are more cost-effective. Smaller islands rely more on diesel generators, but the marginalized communities of these islands benefit from lower capital costs. Jhud M. Aberilla and colleagues [14] conducted research on environmentally sustainable systems for off-grid rural areas. They assessed 21 configurations and found that the (WT+SPV+LA) system, combining wind power, solar PV, and lead-acid batteries, had the least environmental impact across 18 sites. Community diesel generators had the greatest impact, whereas household wind turbines with lead-acid batteries were unfavourable in some scenarios. However, a study by [15] developed a sustainability assessment of energy transition in Cuba. The study analysed the sustainability of migrating from DGs to PV systems in rural Cuba using multidimensional environmental, technical, socioeconomic, and institutional parameters through eight criteria. It further indicated that although diesel generators are the most popular technology for rural electrification in isolated communities in developing countries, different empirical sustainability assessments are needed to assess the long-standing success of RE technology-based projects. Furthermore, a study by the authors [16] focused on off-grid HRES systems in India; HOMER was used for the techno-economic analysis. They found that diesel generator-only systems had a higher net energy cost (1.88 kw/h) and produced 58,362 kg of CO₂ annually. Systems with wind turbines were economically favourable, and the solar/battery/wind/diesel configuration was considered the best for various installation sites and loads. In an evaluation conducted by [17], a hybrid off-grid power system integrating PV, wind, and hydrokinetic energy was assessed for a micro community along the Karun River using HOMER. The study found that the system combining all three energy sources (solar, wind, and hydrokinetic) was the most practical, with the lowest values for NPC, LCOE, and LCOH compared to systems using only one or two energy sources. Interestingly, HOMER couldn't find a viable solution using only solar energy, making the combination of three locally available sources the most economically sensible choice. In addition, a study conducted by [18] on the technical and financial assessment of standalone power systems for different locations in Peru using HOMER. Seven scenarios were evaluated, including various designs for solar PV, wind turbines, and DGs. The ideal size of each configuration was determined by minimizing its related NPC, and the realized ideal systems were then assessed based on other financial metrics and their environmental performance. The assessment showed that the SPV-wind-DG system was the best commercially feasible option for all communities under investigation. A study by [19] investigated the practicability of hybrid solar PV and diesel generator systems (SPV-DG HES) for rural electrification in the Chilubi district of Zambia using HOMER software. The study found

that SPV-DG HES are more economically viable and have lower carbon emissions than stand-alone diesel-generating systems. However, the study also found that the capital costs for PV power installations are higher in Zambia than in other developing countries. Razfar et al. [20] evaluated energy supply for an industrial complex in Iran using HOMER software to simulate and optimize five configurations: solar-only, wind-only, diesel-only, a hybrid (solar, wind, diesel), and a hybrid with diesel priority. The optimized solution consists 20 MW solar panels, two 7.5 MW wind turbines, a 19 MW diesel generator, and 5 MW of battery storage which generated 44.2 % of energy from solar, 36 % from wind, and 19.9 % from diesel, with a total energy production cost of \$0.66/kWh. However, prioritizing diesel reduced the cost to \$0.119/kWh but increased dependency on fossil fuels, which contributed 79.4 % of the total energy. The study also highlighted the environmental impact of diesel-only systems, producing over 61.8 million kg of CO₂ annually, emphasizing the importance of renewable energy integration. Also, Bagdadee and Zhang [21] assessed a hybrid power generation framework for rural electrification in Kuakata, Bangladesh, incorporating solar PV, wind turbines, micro-hydro systems, and diesel generators. The hybrid system achieved an average energy cost of \$0.21/kWh, with solar panels contributing 39.33 %, wind turbines 11.24 %, and micro-hydro providing seasonal stability, generating up to 115,000 kWh monthly during peak water flow. Comparatively, diesel-only systems exhibited higher costs and accounted for 55 % of total CO₂ emissions, underscoring the environmental and economic benefits of hybrid configurations. Moreover, Heidari et al. [22] optimized a hybrid renewable energy system (HRES) for remote communities using HOMER software. The optimal configuration—comprising photovoltaic cells, a diesel generator, a wind turbine, and a battery system—achieved a total net cost of \$180,065.43 and a leveled cost of energy of \$0.32/kWh. This setup reduced operating costs by 53.17 % and carbon dioxide emissions by 60.46 %, while ensuring 100 % load demand fulfilment, highlighting HRES's potential for cost-effective, sustainable, and reliable rural electrification. Similarly, Pujari et al. [23] conducted a study on hybrid renewable energy systems (HRES) in rural India, focusing on the techno-economic feasibility of integrating solar photovoltaic (PV) and wind turbine (WT) technologies. Using HOMER software, they optimized the system configuration to minimize the net present cost (NPC) and cost of energy (COE). Their optimal system consists a 400-kW solar PV array, a 100-kW wind turbine, and a battery storage system, achieving an NPC of \$26.8 million and a COE of \$4.32/kWh. The study demonstrated that the HRES could satisfy 94 % of the community's daily load demand, significantly enhancing energy reliability while promoting sustainable energy practices.

The existing body of literature highlights substantial progress in the design and optimization of hybrid renewable energy systems (HRES) globally, yet several gaps remain. Many studies focus on specific configurations of hybrid energy systems, such as solar PV, wind turbines, and diesel generators, assessing their economic feasibility and environmental performance. However, they often overlook localized factors like real-time energy demand variations and the integration of advanced forecasting techniques such as machine learning models. For instance, while studies like [13,14], and [16] demonstrate the technical and economic viability of hybrid systems in various settings, they rely primarily on traditional statistical methods for energy demand forecasting. These approaches may not adequately capture the complex and nonlinear relationships in energy consumption data, leading to suboptimal system designs. Furthermore, most studies concentrate on environmental and economic factors but do not sufficiently explore the role of predictive analytics in enhancing system efficiency and reliability. Machine learning, with its ability to improve prediction accuracy, remains underexplored in the optimization of hybrid energy systems. Additionally, there is limited focus on standalone hybrid systems tailored specifically for rural electrification in developing countries like Ghana, where unique socio-economic and infrastructural challenges exist. Research comparing hybrid energy systems to grid-extension

alternatives is also scarce, particularly regarding the electric distance limit (EDL) or breakeven distance for cost-effectiveness.

This present study addresses these gaps by integrating machine learning models into the design and optimization process of hybrid energy systems. By leveraging machine learning, the study enhances the accuracy of energy consumption predictions, enabling more reliable and efficient system configurations. This research explores the feasibility and comparative analysis of standalone hybrid energy systems for rural electrification in Ghana. Using HOMER Pro software, the study integrates solar PV, diesel generators, and battery storage to develop sustainable energy solutions. It assesses the technical, economic, and environmental performance of various system configurations to determine the most efficient and optimal solution. In addition, the study extends its analysis by comparing the proposed hybrid systems with grid-extension alternatives, specifically examining the EDL for rural areas. This comprehensive assessment provides insights into the suitability of standalone hybrid systems versus grid extension for remote communities. By addressing these gaps, the research contributes to the broader goal of universal electrification in Ghana and aligns with Sustainable Development Goal 7 (SDG-7), offering practical solutions for policymakers, energy planners, and stakeholders.

2. Materials and methods

The Methodology used in this study is as follows.

- ✓ Energy Resource assessment of the area and load data collection
- ✓ Load assessment and system sizing.
- ✓ Energy Consumption Prediction Using Machine Learning Models
- ✓ System Simulation with HOMER
- ✓ Economic, Technical and Environmental Analysis
- ✓ Comparison with optimal system with grid-connected alternatives.

Potential RE resources were analyzed using climate data obtained from the National Aeronautics and Space Association (NASA) climate database. These data, integrated into the HOMER software program, facilitated the assessment of the village's renewable energy potential for generating electricity.

The load assessment phase of the study involved collecting load data for the village to determine its approximate electricity requirements. This was accomplished through the administration of a survey questionnaire, which allowed researchers to gather information directly from the residents. By analyzing the survey results, the electrical load demands of the village could be estimated, providing valuable insights into the amount of power needed to reach the community's energy requirements. To enhance the accuracy of these estimations and improve system design, machine learning models were integrated into the process. These models were utilized to predict energy consumption patterns, enabling more reliable and precise system configurations tailored to the community's needs.

HOMER was used to simulate four different configurations of electricity generation systems (alternatives), based on the available RE resources in the area.

- ✓ Alternative A (AA) – DG only (Base case)
- ✓ Alternative B (AB) – SPV + DG
- ✓ Alternative C (AC) – SPV + Battery Storage
- ✓ Alternative D(AD) – SPV+ DG + Battery Storage

A comprehensive comparative evaluation was performed to assess the technical, economic, and environmental performance of the four different models for village electrification. The objective was to identify the most viable option that minimizes the total project cost, maximizes renewable energy utilization, and minimizes GHG emissions during operation. The HOMER software was used to simulate and examine the performance of each model. A sensitivity analysis was also conducted on

the most feasible system to examine the impact of economic inputs, such as inflation, discount rate interest rate, and diesel fuel price, on project costs. Key cost parameters, including the NPC, CC, and LCOE, were considered in the project cost analysis. The study also analyzed and compared the optimal solution with grid connection alternatives for the area.

2.1. Hybrid energy system component

A hybrid system consists of several technical components that work together to improve reliability. These technical components work in harmony to optimize the utilization of renewable solar energy, provide backup power through the diesel generator, and store excess energy in batteries for later use. The system's design and configuration depend on the specific requirements, load profile, and energy needs of the application or location where it is implemented.

2.1.1. Solar PV module

SPV panels are fabricated using semiconductor materials that capture the sunlight energy and transform it into electrical energy [24]. These panels are installed to capture solar energy and generate electricity. SPV panels are a crucial component of a hybrid system, as they harness solar energy and convert it into usable electricity. SPV panels are at the core of a hybrid system, providing a dependable and renewable source of electricity. Their installation and integration require careful planning, considering factors such as solar irradiation, site conditions, and load requirements. Among renewable energy sources, solar energy can be considered one of the best natural resources for protecting the environment and reducing energy shortages given the rising demand and consumption of energy in today's world. As solar technology continues to advance, SPV panels are becoming more efficient, cost-effective, and accessible, making them an increasingly popular choice for generating clean and sustainable energy [25,26].

2.1.2. Inverter/Converter

The inverter/converter is an important element of any solar panel system, as its primary function is to transform the Direct Current (DC) electricity produced by SPV modules into Alternating Current (AC) which is suitable for powering various appliances and lighting systems. It also converts AC electricity from the diesel generator into DC electricity for charging the batteries. This is a critical component in a hybrid system, enabling the efficient and effective integration of various energy sources and facilitating the optimal utilization of electricity. It ensures the compatibility, control, and management of AC and DC power, contributing to the hybrid system's overall performance, reliability, and sustainability [27].

2.1.3. Battery storage

Battery storage plays a vital role in a hybrid system by storing surplus energy produced by the SPV panels. It serves as a valuable resource for managing fluctuations in solar energy generation, ensuring a steady power supply during low sunlight conditions or when the diesel generator is not operational. Additionally, battery storage enhances system efficiency, stability, and flexibility, enabling load shifting and demand management. By harnessing stored electricity, battery storage optimizes the utilization of RE sources, reduces GHG emissions, and contributes to a sustainable and reliable energy solution [28–30].

2.1.4. Diesel generator

The diesel generator is an essential component of a hybrid system, providing backup power when solar energy is limited or unavailable. It operates by burning diesel fuel in an engine, which in turn drives an alternator to produce electricity. The diesel generator acts as a reliable power source, ensuring uninterrupted electricity supply during times of low solar irradiation or high energy demand. Although it relies on non-renewable fuel, the diesel generator plays a crucial role in maintaining

system reliability and meeting electricity needs in HESs [31].

2.2. Study area

The study was conducted in Dupare village, located in the Wa East district of the Upper West region of Ghana, and lies along the latitude and longitude of 9°58'19.0" "N 2°13'03.0" W, respectively (Fig. 1). The district capital is Funi and the distance between Dupare and Funi is approximately 89 km. According to the 2021 population and housing census of Ghana, the population of Dupare Village stood at 1425 people with approximately 200 households and is occupied by a land area of approximately 259,730 m² [32]. The main economic activities of the area are agriculture, such as crop farming and livestock rearing. The population of the area was categorized as 41 % of those aged < 15 years and 59 % of those aged ≥ 15 years. The census data obtained from the district assembly indicated that, 76 % of the population age 15 years and older are economically active while 24 are not economically active. Within the economically active group, 98 % of respondents occupied both the formal and informal sectors, leaving 2 % without employment. Census data obtained from the district assembly indicated that 76 % of the population aged 15 years and older were economically active, while 24 were not economically active. The region has a tropical equatorial climate with a single rainfall season from May to October, which is dominant in that area. Throughout the year, temperatures remained consistently high, with the highest temperatures occurring in March and April, reaching 42 °C. In contrast, the lowest temperatures occurred in December and January, dropping to 22 °C. The area also experiences the phenomenon of harmattan, which occurs from November to April and is characterized by cold and dry weather with occasional haze [33].

2.3. Electricity access

The village is an off-grid area, approximately 14 km from the closest national grid connection. The primary energy needs of the village are electricity and cooking, whereas they rely heavily on biomass (firewood and charcoal) for cooking purposes and small lanterns and torchlights fueled by dry cells for lighting purposes. Few elite households have solar panels on their roofs that power basic direct-current (DC) loads. Most mobile phones of the community residents get their mobile phones charged in the closest village with grid access, which most complain of losing their phones through this arrangement. The village falls under tier 1 of the UN multi-framework for energy access, where small lanterns and torchlights with batteries or photovoltaics are used to meet the lighting needs of households.

2.4. Energy resource assessment

This study relied on input variables such as the solar irradiance of the location to examine the electricity output of the HES. There are different approaches to obtaining such data: one is to collect ground data using on-site meteorological instruments, and the other approach utilizes the HOMER software, which provides access to satellite meteorological data maintained by NASA. This study employed the latter, where data from the NASA database of the selected location were used to eliminate the need for ground data in the simulation.

2.4.1. Solar irradiance

Solar irradiance is a measure of the power of sunlight per unit area and is measured in watts per square meter (W/m²) [36]. This is an important parameter for assessing solar energy resources at any location. With an estimated annual average sunshine of 5.75 h, solar energy is a viable source of power for the site. Higher levels of solar irradiance are experienced from February to May and October, while the least recorded level of irradiance is recorded in July–August, when rainfall is at its peak. Fig. 2(a) shows the average monthly solar irradiance at this location.

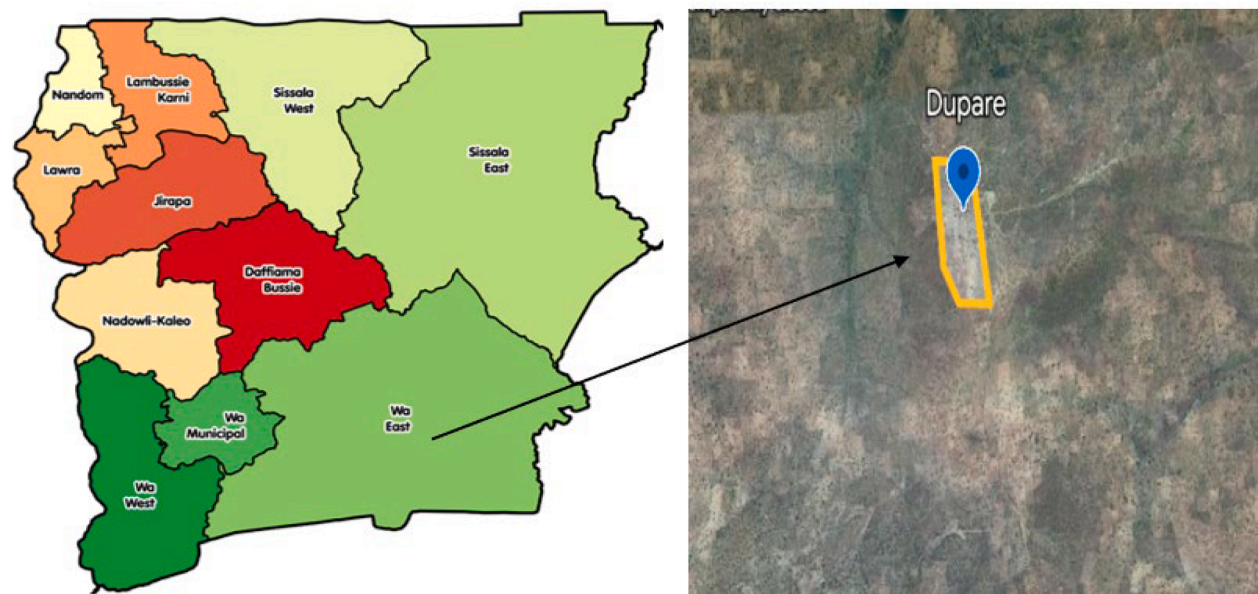


Fig. 1. Map of case study region and google earth view of study area [34,35].

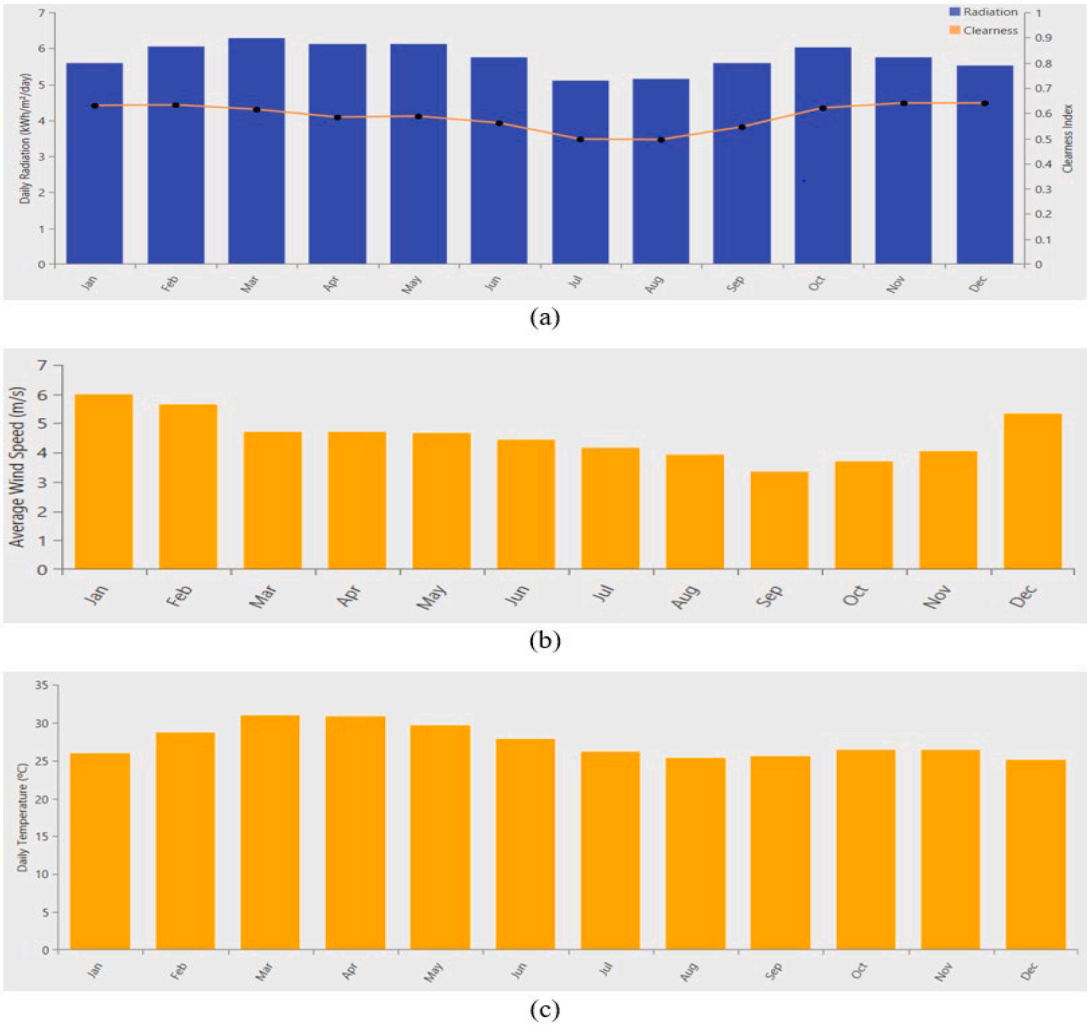


Fig. 2. (a) Average Monthly Solar Irradiance of Study Area (b) Average Monthly Wind Speed at Study Area (c) Monthly Average Air Temperature of the Study Area [25].

2.4.2. Wind speed

The average wind speed of the area measured 50 m above the ground is 4.55 m/s which indicates low feasibility for wind energy generation at that location. Fig. 2(b) shows the average monthly wind speed in the area.

2.4.3. Temperature

The average annual temperature of the area is 27.39 °C with the peak average monthly period being the dry season (February-May). The variation depends on the time and season of the year. Fig. 2(c) shows the monthly average temperature at this location.

2.5. Electrical load estimation

Electrical load estimation involves determining the energy consumption needs of various appliances and equipment within a specific setting such as households, shops, or schools. A sample size of 80 persons comprising 80 households (40 %) and business owners was interviewed to ascertain electricity requirements. The electrical load requirements of community facilities such as schools, mosques, churches, and health centres were also obtained with the help of the leadership of those facilities. It considers factors such as the number of rooms, family size, and the types of electrical devices present. Standard assumptions were applied to approximate the load power ratings, ensuring adequate consideration of nonenergy-saving appliances. This estimation is crucial for designing efficient and sustainable energy systems that meet the specific needs of a location.

2.5.1. Load categorization

To estimate the load, the load demand of the village was categorized into three categories: residential, commercial, and community loads, as shown in Table 1.

2.5.2. Load requirement

The projected energy consumption for the area was determined using standard assumptions concerning the type of load, electrical power demand, and duration of usage, which were all assessed through a survey. The following assumptions were made for this study:

All equipment were sized using standard wattage ratings typically associated with these appliances. A load assessment was performed by considering the living and proposed consumption patterns of the occupants. Load assessments were categorized into three categories: residential loads (households), commercial loads (shops), and communities (health facilities, schools, mosques, and churches).

The assumed electrical loads for a standard household include five (5) light bulbs (indoor and outdoor), a fan, a television (TV), a radio, a min-fridge, and two phones. The assumed electrical load for a shop includes three (3) light bulbs (indoor and outdoor), a fan, TV, radio, refrigerator, and two phones. The community has a school, health facility, mosque, and church whose loads are equally sized according to their needs.

The load assessment of the village was computed using excel sheet. Electrical energy was computed as follows: electrical energy = electrical power of appliances (Watts) × time of use (h).

A summary of the computed daily electricity needs of the village per

Table 1
Summary of load categorization.

Category	Type of load	Quantity
Residential Loads	Households	200
Commercial Loads	Shops	6
Community Loads	School	1
	Health facility	1
	Mosque	1
	Church	1

categorization is presented in Table 2.

2.6. Energy consumption prediction using machine learning models

The development and evaluation of various machine learning models to predict energy consumption based on features such as 'Power Rating (W)', 'Hours of Use in a Day (H)', and 'Quantity' was explored. The models explored include Linear Regression, Decision Tree, and Random Forest Regression. We also compare the performance of these models using metrics such as RMSE, MAE, and R.

The prediction process was initiated by preprocessing the data. In this step, the data was normalized, and some features were selected. After that, each of the models was trained on the dataset. The performance for each model with regard to evaluation metrics like RMSE, MAE, and R² was considered for estimation using cross-validation. Later, the predictions obtained from these models are integrated with HOMER software to simulate hybrid energy system performance in different scenarios. We designed the study around a dataset that included information on the ratings of the various appliances, hours of use per day, and quantity. Features were engineered to capture the relationship between these variables and their impact on overall energy consumption. These included three machine learning models that were developed and compared for their predictive performance. Data preprocessing was done first. Following that, the prediction phase was completed by creating models for each in the dataset and testing the performance of those using cross-validation, RMSE, MAE, and R² metrics. Afterwards, the predictions from these models were put into the HOMER software to simulate the performance of hybrid energy systems under different scenarios.

2.6.1. Data processing a featured engineering

The dataset contains information on household appliances, including their power ratings, usage hours and energy consumption.

2.6.1.1. Feature engineering. New features were created to capture interactions between 'Power Rating' and 'Hours of Use'

$$\text{Energy Per Unit} = \text{Power Rating (W)} * \text{Hours of Use in a Day (H)}$$

$$\text{Energy Efficiency} = \frac{\text{Energy (WH)}}{\text{Power Rating (W)} * \text{Hours of Use in a Day (H)}}$$

2.6.2. Model development

We developed three models: Linear Regression, Decision Tree Regression, and Random Forest Regression.

2.6.2.1. Linear regression algorithm.

1. Input: Dataset $D = \{(X_i, y_i)\}_m$ where $X_i = [\text{Power Rating}_i, \text{Hours of Use}_i]$ and $y_i = \text{EnergyWH}_i$.
2. Initialize: Set initial weights $\beta_0, \beta_1, \beta_2$.
3. Hypothesis: Model the energy consumption as a linear combination of the features:

$$\hat{y}_i = \beta_0 + \beta_1 \times \text{Power Rating}_i + \beta_2 \times \text{Hours of Use}_i \quad (1)$$

Table 2
Summary of daily electricity consumption.

Item	Residential	Commercial	Community	Total
Daily electricity consumption (kwh/day)	307.200	27.708	48.596	383.504

4. Loss Function: Define the mean squared error (MSE) as the loss function:

$$J(\beta_0, \beta_1, \beta_2) = \frac{1}{m} \sum_{i=1}^m (\hat{y}_i - y_i)^2 \quad (2)$$

5. Optimization: Use gradient descent to minimize the loss function:

$$\beta_j := \beta_j - \alpha \frac{\partial}{\partial \beta_j} J(\beta_0, \beta_1, \beta_2) \quad \text{for } j = 0, 1, 2 \quad (3)$$

6. Convergence: Repeat the optimization step until the changes in the loss function are below a defined threshold.

7. Output: The final weights $\beta_0, \beta_1, \beta_2$ used to predict energy consumption.

2.6.2.2. Decision tree regression algorithm.

1. Input: Dataset $D = \{(X_i, y_i)\}_m$ where $X_i = [\text{Power Rating}_i, \text{Hours of Use}_i]$ and $y_i = \text{EnergyWH}_i$.
2. Tree Construction: Recursively partition the data at each node:
 - Choose the feature and split point that minimizes the MSE.
 - Split the data into two subsets.
 - Repeat the process for each child node until a stopping criterion is met (e.g., maximum depth or minimum samples per leaf).
3. Prediction: For a new input X , traverse the tree from the root to a leaf, returning the mean value of the target variable at that leaf.
4. The predicted energy consumption $\hat{y}$ for new inputs.

2.6.2.3. Random forest regression algorithm.

1. Input: Dataset $D = \{(X_i, y_i)\}_m$ where $X_i = [\text{Power Rating}_i, \text{Hours of Use}_i]$ and $y_i = \text{EnergyWH}_i$.
2. Parameters: Number of trees B , number of features m_{try} to consider at each split, maximum depth d_{max} , and minimum samples per leaf n_{min} .
3. Bootstrap Sampling: For each tree b (where $b = 1$ to B):
 - (a) Draw a bootstrap sample D_b from D .
 - (b) Grow a regression tree T_b on D_b by:
 - Randomly selecting m_{try} features at each split.
 - Splitting the node using the feature that minimizes the mean squared error (MSE).
 - Repeat until the maximum depth d_{max} is reached or the node contains fewer than n_{min} samples.
1. Prediction: For a new input X , the prediction $\hat{y}$ is the average of the predictions from all trees:

$$\hat{y} = \frac{1}{B} \sum_{b=1}^B T_b(X) \quad (4)$$

2. Output: The predicted energy consumption $\hat{y}$ for new inputs.

2.6.3. Model evaluation

2.6.3.1. Cross-Validation. We performed K-fold cross-validation to ensure the model's generalization performance. The dataset was split into K subsets, and the model was trained on $K - 1$ subsets and tested on the remaining subset.

$$CV \text{ Score} = \frac{1}{K} \sum_{i=1}^K \text{Performance}(D_i^{\text{test}}, D_{-i}^{\text{train}}) \quad (5)$$

2.6.3.2. Evaluation Metrics. The models were evaluated using the following metrics:

- Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{\frac{1}{m} \sum_{i=1}^m (\hat{y}_i - y_i)^2} \quad (6)$$

- Mean Absolute Error (MAE):

$$MAE = \frac{1}{m} \sum_{i=1}^m |\hat{y}_i - y_i| \quad (7)$$

- R-Squared (R^2):

$$R^2 = 1 - \frac{\sum_{i=1}^m (y_i - \hat{y}_i)^2}{\sum_{i=1}^m (y_i - \bar{y})^2} \quad (8)$$

2.6.4. Model performance comparison

The [table 3](#) below compares the performance of the three models using the evaluation metrics:

2.6.4.1. Interpretation. The Decision Tree model has the lowest RMSE and MAE, indicating that it makes the most accurate predictions on this dataset. It also has the highest R^2 value, suggesting that it explains more variance in the energy consumption than the other models. The Random Forest model performs better than Linear Regression in all metrics, but not as well as the Decision Tree model, which might be due to the specific nature of the dataset, where a simpler model like a single Decision Tree can capture the key relationships effectively. The Linear Regression model, while the simplest, performs the worst in terms of RMSE, MAE, and R^2 , which is expected given that the relationship between the features and the target variable is likely non-linear.

2.6.5. Feature importance analysis

The Random Forest model provides a way to measure the importance of each feature in predicting energy consumption. The importance of a feature is determined by the decrease in the model's performance when the feature is removed. The higher the importance, the more significant the feature is in predicting the target variable.

Table 3
Comparison of Model Performance Metrics.

Metric	Random Forest	Linear Regression	Decision Tree
RMSE	2007.03	2270.68	1838.75
MAE	1095.83	1553.57	781.67
R^2	0.4124	0.2479	0.5068

2.6.5.1. Results and visualization of feature importance. Table 4 shows the importance of each feature in the Random Forest model. Fig. 3 visualizes the importance of each feature. The bar chart provides a clear view of which features contribute the most to predicting energy consumption.

2.6.5.2. Interpretation of feature importance.

- **Power Rating (W):** This feature has the highest importance (0.4286), indicating that it plays the most critical role in predicting energy consumption. This result is expected, as power rating directly impacts the amount of energy an appliance consumes.
- **Quantity:** With an importance of 0.3089, the number of appliances contributes significantly to the overall energy consumption, as having more units naturally increases the energy usage.
- **Hours of Use in a Day (H):** This feature is important at 0.2625, showing that the duration for which the appliances are used is also a crucial factor in determining energy consumption.

2.7. HOMER as optimization software

HOMER, developed by the National Renewable Energy Laboratory (NREL), is versatile optimization software for renewable energy systems. Its powerful engine finds optimal configurations by considering energy demand, resource availability, costs, and constraints. HOMER supports various renewables such as solar and wind energy, and even integrates conventional generators. With user input parameters, HOMER generates detailed reports and visualizations to evaluate energy production, costs, and more. It is widely used in projects such as rural electrification and microgrids, aiding decision-makers in designing efficient and sustainable systems (Fig. 4).

2.8. Simulation parameters

Simulation parameters play a crucial role in the analysis and optimization of energy systems. The key input parameters for SPV systems include the panel efficiency, rating, and type. In the case of converters, parameters such as the efficiency and conversion efficiency are vital. Battery systems require input parameters, such as capacity, efficiency, and depth of discharge. The fuel price, efficiency, and maintenance costs are essential considerations for diesel generation. Economic parameters include interest rate, inflation rate, and discount rate, which impact the financial aspects of the energy system.

2.8.1. Input data for SPV

The SPV module of the HES makes use of the sun's energy during the daytime to generate electricity for consumption and the extra energy utilized to charge the battery component where necessary. The PV panel characteristics and study parameters used are summarized in Table 5.

2.8.2. Input data for battery storage system

Energy is stored in batteries and serves as backup power when needed. With the HES, the BESS is charged during the day by the SPV System and releases power for consumption during the night or on cloudy days. The parameters of the batteries used are listed in Table 6.

Table 4
Feature Importance in Random Forest Model.

Feature	Importance
Power Rating (W)	0.4286
Quantity	0.3089
Hours of Use in a Day (H)	0.2625

2.8.3. Input data for converter

Converters play a key role in energy systems by facilitating the electrical power conversion between different forms. In the context of RE systems, such as SPV or wind, converters are essential to transform the generated DC into AC electricity that can be used to power homes, industries, and other electrical appliances. HOMER simulates the converter required for this study with the parameters listed in Table 7.

2.8.4. Diesel generator parameters

The reason for a DG in an HES is to supply backup power in the inability of the SPV and battery storage system to do so based on the set priority. The HOMER simulation software sized a diesel generator that matched the load requirement in this study with the parameters summarized in Table 8.

2.8.5. Economic analysis parameters for sensitivity analysis

HOMER simulation software requires economic parameters such as inflation and discount rate for the purpose of this study. The central bank of Ghana as of July 2023 quoted 25.9957 % as the discount rate [37] and the year-on-year inflation rate as of June 2023, as reported by [38], to be 42.5 %, with expected inflation rate forecasted by [39] to be 10 % in 2024 and 8 % in 2025.

Sensitivity analysis is also known as “what if” analysis and is usually performed to determine how an input variable affects an output variable. For this purpose, input variables such as inflation rate, cost of fuel, and discount rate are chosen. On the other hand, output variables such as NPC, CC, LCOE, and O&M costs are observed. The economic parameters for the sensitivity analysis are summarized in Table 9.

2.9. Model equations

2.9.1. Levelized Cost of Energy (LCOE)

$$LCOE = \frac{\text{sum of costs over lifetime}}{\text{sum of electrical energy produced over lifetime}} = \frac{\sum_{x=1}^n \frac{I_x + M_x + F_x}{(1+r)^x}}{\sum_{x=1}^n \frac{E_x}{(1+r)^x}} \quad (9)$$

[40]

I_x = investment costs in the year x

M_x = O&M cost in the year x

F_x = Costs related to fuel consumption in the year x

E_x = Amount of electricity produced in the year x

r = Discount rate

n = Anticipated lifespan of the system or power station

2.9.2. Annualized cost

$$C_{ann} = CRF(i, N) * CNPC \quad (10)$$

[41]

Where:

C_{ann} = Annualized cost [\$]

$CRF()$ = Capital Recovery factor function

N = Project lifetime[years]

i = The annual real discount rate [%]

C_{NPC} = Net Present Cost [\$]

$$CRF(i, N) = \frac{i(1+i)^N}{(1+i)^N - 1} \quad (11)$$

[41]

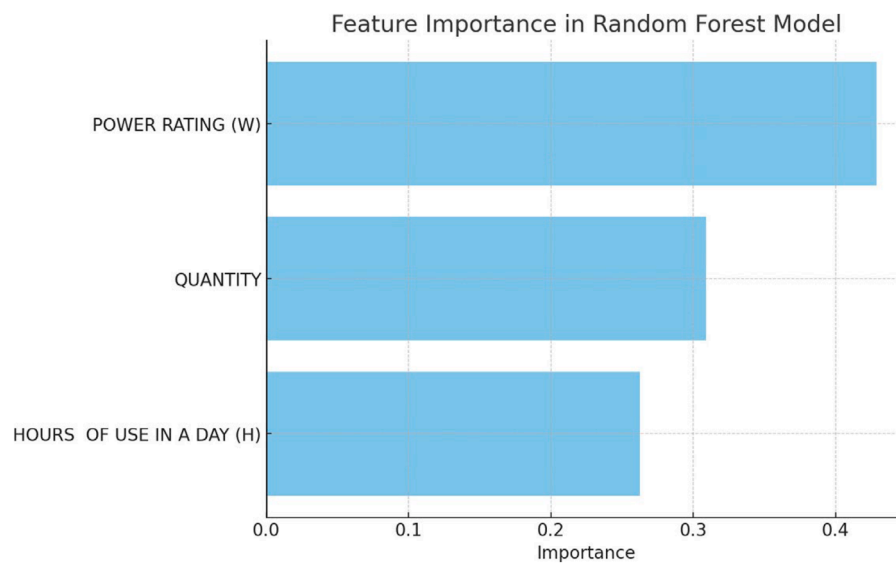


Fig. 3. Feature Importance in Random Forest Model.

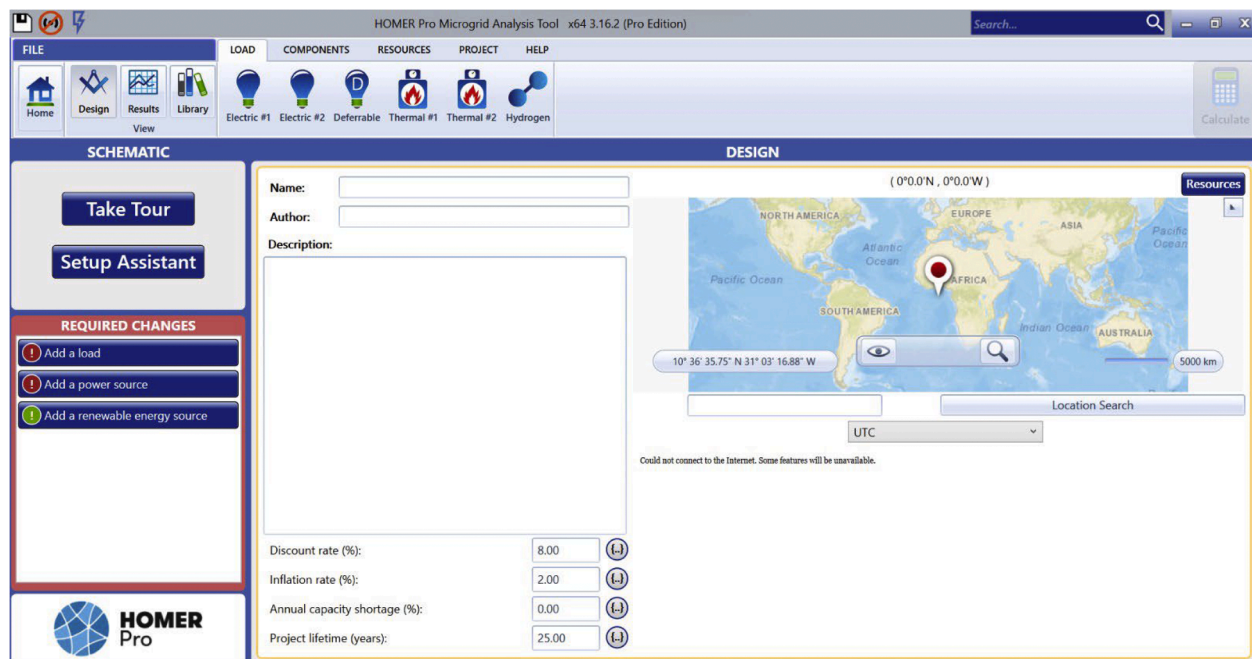


Fig. 4. User Interface of HOMER Software.

Table 5
SPV Module Parameters.

PV Module Specifications	
Manufacturer	JA Solar
Item no.	JAM66S-30-500-MR-MC4
Type	Monocrystalline
Rated Max. Power (W)	500
Voltage at Pmax (V)	38.35
Current at Pmax (I)	13.04
Efficiency (%)	21.1
Derating factor (%)	80
Lifetime (Years)	25
Capital Cost per kW (\$)	500
Yearly O&M Cost per kW (\$/year)	10

Table 6
Battery Parameters.

Battery Specifications	
Manufacturer	Generic Lithium-ion
Type	Lithium-ion
Rated Voltage (V)	6
Battery capacity (Ah)	167
Nominal Capacity (kWh)	1
Efficiency (%)	90
State of Charge (Initial) (%)	100
State of Charge (Minimum) (%)	20
Throughput (kWh)	3000
Lifetime (years)	15
Unit Cost (\$)	200

Table 7

Parameters of Converter.

Converter Parameters	
Manufacturer	Generic Converter
Type	Hybrid System Converter
Efficiency (%)	95
Battery Voltage (VDC)	48
Lifetime (year)	15
Capital Cost (\$/kW)	200

Table 8

Parameters of Diesel Generator.

Diesel Generator Parameters	
Manufacturer	Perkins
Fuel Type	Diesel
Capital Cost (\$/kW)	400
Minimum Load ratio (%)	25
Replacement Cost (\$)	400
Cost of diesel (\$/liter)	1.06
O&M Cost (\$/hr)	0.05
Lifespan (hours)	15,000

Table 9

Economic Parameters and Sensitivity Input Variables.

Economic parameter and Input variables of Sensitivity Analysis		
Parameter	Rate	Input values for sensitivity
Current Inflation rate (%)	42.5	n/a
Expected Inflation rate (%)	10	30,15,8
Discount rate (%)	25.9957	28,14,7
Diesel price (\$)	\$1.06	\$1.2, \$0.90, \$0.95

$$NPC = \frac{\text{Cash flow}}{(1+i)^k} - \text{initial investment} \quad (12)$$

[42]

Where:

 i = Discount rate k = Total number of time Period

2.9.3. Salvage value

$$S = K(1-r)^x \quad (13)$$

[43]

Where:

 K = Original price r = Depreciation rate x = age in years S = Salvage value

2.9.4. Real discount rate

$$i = \frac{(a-f)}{(1+f)} \quad (14)$$

[44]

Where:

 a = nominal discount f = expected inflation rate i = real discount

The real power (P_{PV}) of the PV panel

The real power (P_{PV}) of the PV panel under real operation and cli-

matic conditions is

$$P_{PV} = P_{STC} \left[1 + \beta_p (T_{cell} - T_{STC}) \right] \frac{H_t}{H_{STC}} \quad (15)$$

Where: T_{STC} and H_{STC} are standard test conditions of temperature [$^{\circ}\text{C}$] and solar radiation [Watt/m^2] respectively, β_p is the power temperature coefficient [45].

$$T_{cell} = T_{\infty} + 7.8 \times 10^{-2} H_t \quad (16)$$

Where: H_t is the solar radiation intensity [Watt/m^2], T_{∞} and T_{cell} are the ambient air and the surface cell temperatures [$^{\circ}\text{C}$], respectively [46].

3. Results and discussion

This section presents the performance of the simulated hybrid energy system configurations, focusing on their ability to meet the village's energy needs while keeping costs low, maximizing the use of renewable energy, and minimizing greenhouse gas emissions. The analysis, conducted using HOMER software, evaluates each system's technical, economic, and environmental viability. Additionally, sensitivity analyses explore how changes in factors like inflation rates, diesel prices, and discount rates impact the systems' overall performance. The aim is to identify the most practical and sustainable solution for rural electrification in Ghana, balancing affordability with environmental benefits.

3.1. Simulation options

The simulations were conducted for four distinct scenarios as presented in table 10 whiles Fig. 5 shows the schematics of the different scenarios.

3.2. Comparison of the alternatives

The simulated and optimized alternatives presented by HOMER have different component sizes for various configurations. This is because the systems were designed to satisfy the energy load of the village at all times. The diesel generator (DG) is 80 kW for all the different configurations because it must be able to meet the continuous power demand and peak load at any time. The SPV system was designed to overcome the challenges of climate change (cloud cover) and satisfy the load requirements during the day. HOMER considers all of these factors when sizing the components of each configuration, which is why some systems, such as the solar PV, converter, and batteries, have different sizes, whereas the DG has the same size in all configurations.

A summary of the simulated and optimized parameters of the various alternatives is presented in Table 11 below.

Following the simulation and optimization process, Alternative A (AB), which utilizes only DG as the sole power generator, exhibits the highest NPC of \$729,515 and the least CC OF \$32,000 among the four alternatives. Alternative B (AB) showed the second-highest NPC and CC at \$698,812 and \$177,438, respectively. Alternative C (AC) boasts the third-highest NPC of \$308,001 and the highest CC of \$245,474, whereas alternative D (AD) exhibits the lowest NPC and third-highest CC of \$244,937 and \$166,970.

Regarding running costs, including O&M, fuel, and replacement costs, Alternative A had the highest expenditure of \$697,943, as

Table 10

Different Configurations of Scenarios.

Option	Configuration
Alternative A (AA)	DG only
Alternative B (AB)	SPV + DG
Alternative C (AC)	SPV + Battery Storage
Alternative D(AD)	SPV + DG + Battery Storage

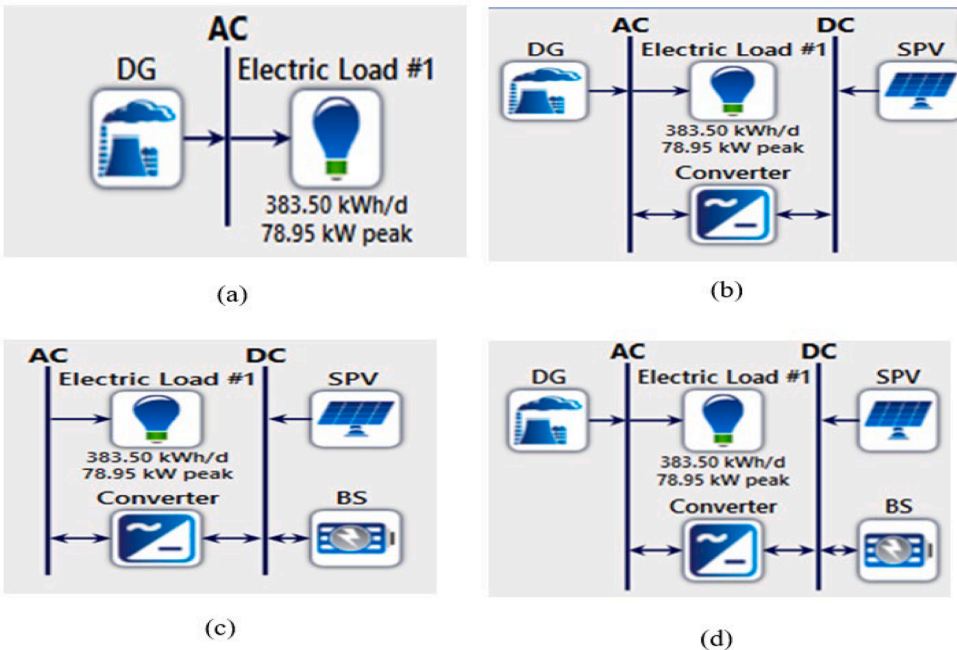


Fig. 5. Schematic of (a) AA (b) AB (c) AC (d) AD.

Table 11
A summary of results of four alternatives.

	Alternative A (AA)	Alternative B (AB)	Alternative C (AC)	Alternative D (AD)
Architecture	DG Only	SPV+DG	SPV+BS	SPV+BS+DG
				
Net Present Cost (NPC) (\$)	729,515	698,812	308,001	244,937
Capital Cost (CC) (\$)	32,000	177,438	245,474	166,970
Replacement Cost (\$)	117,519	82,129	14,535	11,958
O&M (\$)	93,127	85,825	49,239	36,239
Fuel Cost (\$)	487,297	353,954	0	30,358
LCOE(USD/kWh)	0.9346	0.7513	0.331	0.2633
Total electricity generation (kWh/year)	195,912	614,607	458,578	202,817
Excess electricity generation (kWh/year)	55,933	472,386	303,499	48,273
Renewable fraction (%)	0	0	100	91.5
Fuel Consumption (L/year)	69,188	46,545	0	4310
GHG emission (kg/year)	182,794.23	122,971.11	0	11,388

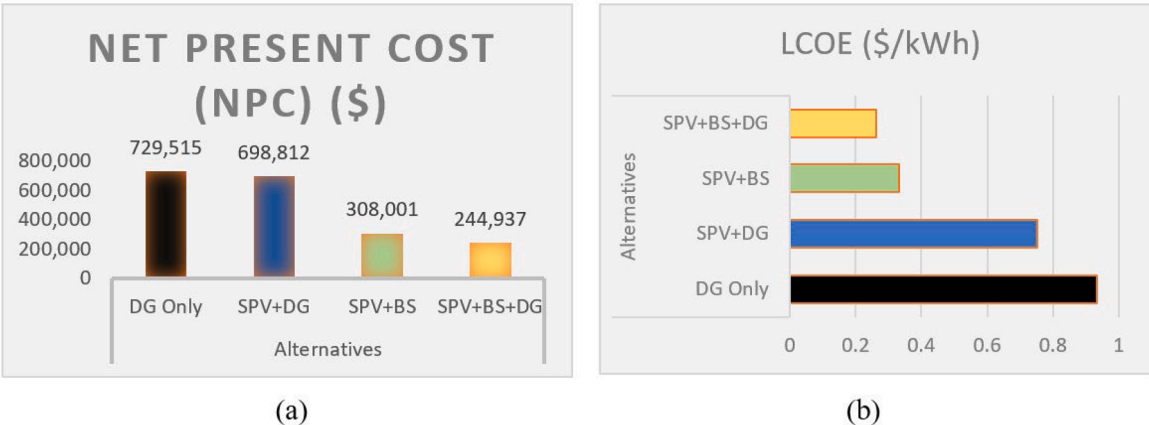


Fig. 6. Comparison of the various alternatives on (a) NPC (b) LCOE.

indicated in the table above. Similarly, Alternative B also exhibits a high running cost of \$521,908, owing to the frequent use of DG. However, Alternative C boasts a relatively low running cost of \$63,774 owing to the non-consumption of fuel and minimal maintenance of the SPV and BS systems. For Alternative D, where the DG serves as a backup power supply with a shorter runtime of 574 h/year compared to Alternatives A and B, the total fuel expenditure of \$30,358 and running cost of \$78,555 over a 25-year period were the lowest among the three alternatives. Figs. 6(a) and (b) present comparisons of the NPC and LCOE of various alternatives, respectively.

The LCOE for the four alternatives, namely AA, AB, AC, and AD, are \$0.9346/kWh, \$0.7513/kWh, \$0.331/kWh, and \$0.2633/kWh, respectively. The results reveal that AD stands out as the alternative with the most affordable LCOE among the four.

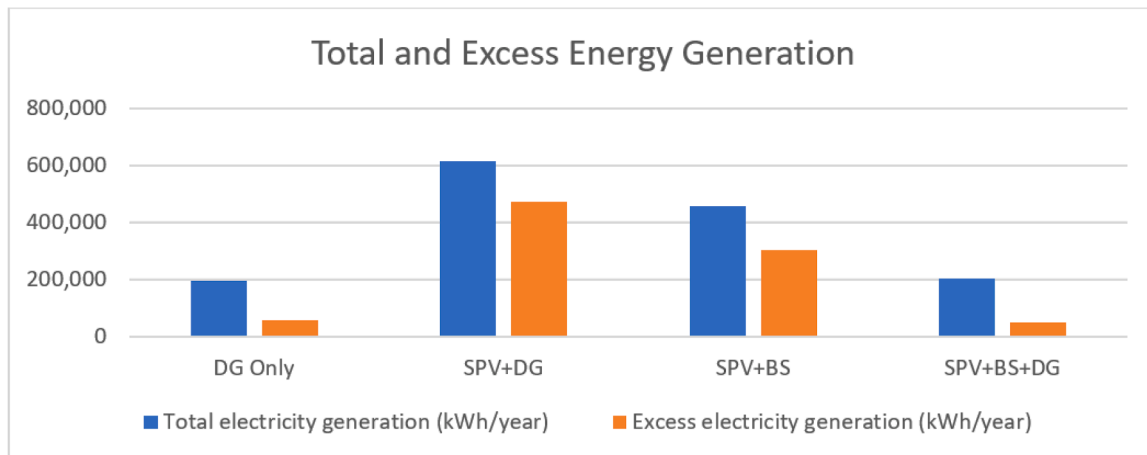
The estimated village load demand indicated a total load of 139,979 kWh/year, which must be met annually. In AA, the total electricity production was 195,912 kWh/year, resulting in 55,933 kWh/year (28.6 %) of the excess electricity. AB generates a cumulative amount of electricity of 614,607 kWh/year, with excess production of 472,386 kWh/year (77 %). AC generates 458,578 kWh/year of electricity, with excess production of 303,499 kWh/year (66 %). Finally, AD generated 202,817 kWh/year of electricity, with a lower excess of 48,273 kWh/year (23.8 %). This results in the AD system having the second-lowest electricity generation and the least surplus energy produced among the four cases. The total and excess energy generation of the various alternatives are depicted in Fig. 7(a).

Among the four scenarios, it is evident that the amount of fuel consumed is directly proportional to the level of GHG emissions. Alternative C, which has no fuel consumption, is expected to produce no emissions over its projected lifetime. In contrast, Alternative A consumes 69,188 L of diesel fuel annually, resulting in 182,794 kg of GHG emissions. For Alternatives B and D, the diesel fuel consumption is 46,545 L and 4,310 L, respectively, leading to GHG emissions of 122,971 kg and 11,388 kg, respectively. Alternative D showed significant emission savings of 171,406 kg and 111,583 kg compared to Alternatives A and B, respectively. The fuel consumption and corresponding GHG emissions are presented in Fig. 7(b).

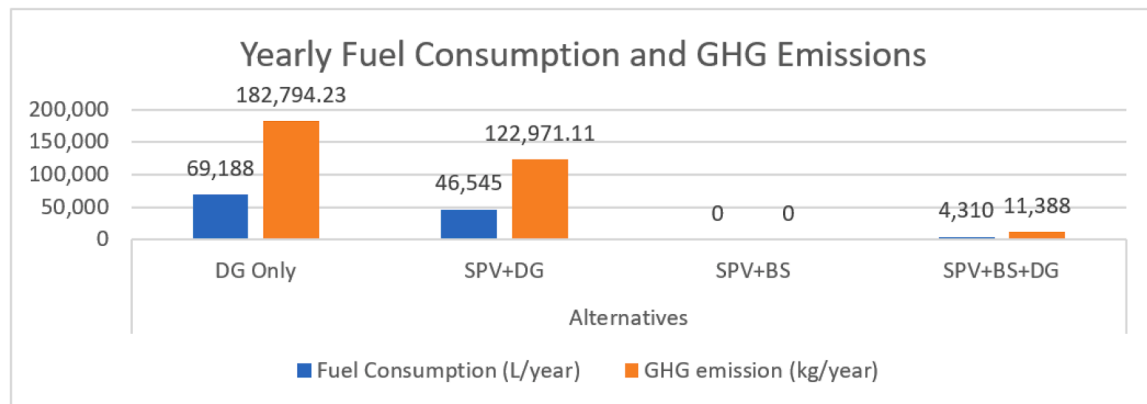
3.3. The recommended solution

The HOMER software was utilized for optimization to evaluate the technical, economic, and environmental viability of a hybrid power generation system, ensuring a dependable electricity supply for the village. A comparative analysis of the four distinct alternatives explored in this study identified the most favourable solution for the village's electricity supply, considering specific economic, technical, and environmental variables. Among the four alternatives, alternative D emerged as the proposed system, relying on an SPV array as the main electricity source and BESS and DG as backup power supplies. The proposed system has the following advantages:

- It exhibited the lowest NPC and LCOE values among all alternatives.



(a)



(b)

Fig. 7. Comparison of alternatives on (a) Total and Excess Electricity generation (b) Fuel Consumption and GHG emissions.

- It maximizes the use of renewable energy resources while minimizing GHG emissions, fully catering to the village's electrical load requirements with minimal electricity waste.

Table 12 presents the architecture of the proposed system, while Fig. 8 and 9 shows the performance of the optimal system and its components respectively.

3.4. Excess energy utilization for socio-economic and community development

Among the various systems analyzed, AD as the optimized configuration produced the least excess energy generation at 48,273 kWh/year, accounting for 23.8 % of the energy generated by the system. Insights gathered from community engagement underscore the economic limitations imposed by the absence of electricity. The introduction of access to electricity could act as a catalyst for local economic activities, fostering increased business opportunities. Surplus annual energy output plays a pivotal role in this transformation. This excess energy reserve could also be harnessed to accommodate potential increases in household energy needs or future expansion. Additionally, it presents an avenue to power essential community assets such as water pumps, further enhancing the overall well-being of the community. This could be hinged upon by understanding the load profile and establishing effective energy-management practices.

3.5. Sensitivity analysis

Sensitivity analysis was done to evaluate the economic performance of the optimal hybrid energy system considering inflation rates, fuel prices and discount rates.

For the case of inflation as depicted in Fig. 10, the baseline value of 10 % yielded an NPC of \$244,936 and an LCOE of \$0.263/kWh. A decrease to 8 % reduced NPC by 4 % but increased LCOE by 8 %, while an increase to 15 % raised NPC by 12.5 % and reduced LCOE by 19 %. At an inflation rate of 30 %, NPC rose significantly by 51 %, whereas LCOE dropped by 65 %, highlighting the strong influence of inflation on system economics.

Similarly, fuel price variations affected economic outputs. At the baseline diesel price of \$1.06/litre, NPC and LCOE were \$244,936 and \$0.263/kWh as shown in Fig. 11. Lowering the price to \$0.90/litre decreased both metrics by approximately 1.9 %, while increasing the price to \$1.20/litre raised NPC and LCOE by 1.5 %. These modest changes indicate a relatively low sensitivity of the system to fuel price fluctuations.

For discount rates, a baseline of 26 % resulted in an NPC of \$244,936 and an LCOE of \$0.263/kWh as depicted in Fig. 12. Reducing the discount rate to 7 % dramatically increased NPC by 50 % but decreased LCOE by 63 %. Conversely, increasing the discount rate to 28 % lowered NPC by 3.6 % but raised LCOE by 6.7 %. These findings demonstrate that the discount rate is a critical factor influencing the financial viability of the hybrid energy system.

3.6. Comparison of proposed system with grid connected

The selection of a system to cater to an off-grid load is aligned with previous research by [47,48]. However, certain differences exist, such as

the cost of components and load size, which might account for variations in figures, particularly in metrics such as the Levelized Cost of Energy (LCOE). Notably, earlier studies have indicated the superiority of hybrid systems in meeting the load demands of rural areas distant from the national grid, as mentioned in the overview of this work. In addition to the variation in system types, it is exciting to see that the LCOE in the present study (\$0.263/kWh) is lower than the findings in [48] (\$0.281/kWh) and [49] (\$0.399/kWh) concerning rural electrification in Ghana. This implies that the recent changes in solar PV costs and the surge in fuel prices and inflation rates during the COVID-19 pandemic have influenced the outcomes of past studies. The average electricity end-user tariff as approved PURC by effective June 2023 was GHC1.264 (\$0.115) for grid-supplied electricity. Thus, the LCOE derived from the standalone DG/SPV/BS system remains relatively high compared with the LCOE for grid-connected remote communities.

The break-even distance or Electric Distance Limit (EDL) for the grid extension distance corresponds to the distance at which the NPC of extending the grid equals the NPC of the standalone system. According to [50], the average cost for extending the grid in sub-Saharan Africa is approximately \$25,000 per kilometre for the 11 kV line. By adjusting this estimate by 30 % to account for differences in terrain and possible inflation resulting in \$32,500, the breakeven distance or Electric Distance Limit (EDL) for the proposed system is 7.5 km which is less than the distance of most rural and island communities in Ghana from the grid. The distance from the case study area to the nearest grid point is 14 km. Extending the grid to rural areas is financially impractical because of the lack of transmission lines. While the cost associated with grid connection continues to rise, the fall in prices of RE products in the market strengthens the argument for rural electrification through standalone HES.

4. Conclusions

The present study provides an in-depth feasibility design and comparative evaluation of a standalone hybrid energy system for rural electrification in Ghana using Hybrid Optimization of Multiple Energy Resources focusing on the development and evaluation of various machine learning models in order to estimate the power consumption based on features such as power rating, hours of use per day, and number of appliances. The models tested included linear regression, decision tree, and random forest regression. The performances were compared based on RMSE, MAE, and R^2 . These predictions were then fed into the HOMER software to simulate the performance of such a hybrid energy system under different scenarios. The study examines the design and multiple configurations of different alternative energy systems using the data required with respect to the study area. Alternative D (AD) has emerged as the best optimized and recommended solution. It consists of a 112 kW SPV array, 80 kW DG, 342 kWh BS and a 54-kW converter. The LCOE obtained from the optimal system was \$0.0263/kWh, which was the lowest among the alternatives. The optimal configuration had a renewable fraction of 91.5 %, with less excess electricity (23.8 %). The NPC system was also obtained as the least among the alternatives, reaching \$244,937. The optimal system demonstrated an exemplary balance between renewable energy utilization, cost efficiency, and reliability, making it the most suitable choice for providing electricity to rural communities. The integration of these models allowed for a more robust analysis, ensuring that the proposed hybrid systems could meet the energy demands of rural communities effectively and sustainably.

The LCOE of the optimal system in this study is \$0.263/kWh, which is notably lower compared to the LCOE figures reported in [48] (\$0.281/kWh) and [49] (\$0.399/kWh) for rural electrification in Ghana. This disparity in the LCOE values indicates that recent developments in the solar photovoltaic (PV) industry, such as reductions in PV module costs and improved system efficiencies, have played a significant role in influencing the outcomes of past studies. Additionally, the unprecedented circumstances surrounding the COVID-19 pandemic

Table 12
Summary of Proposed HES Architecture.

Component	Specification
Diesel Generator	80 kW
SPV Array	112 kW
Battery Storage	342 kWh
Converter	54 kW

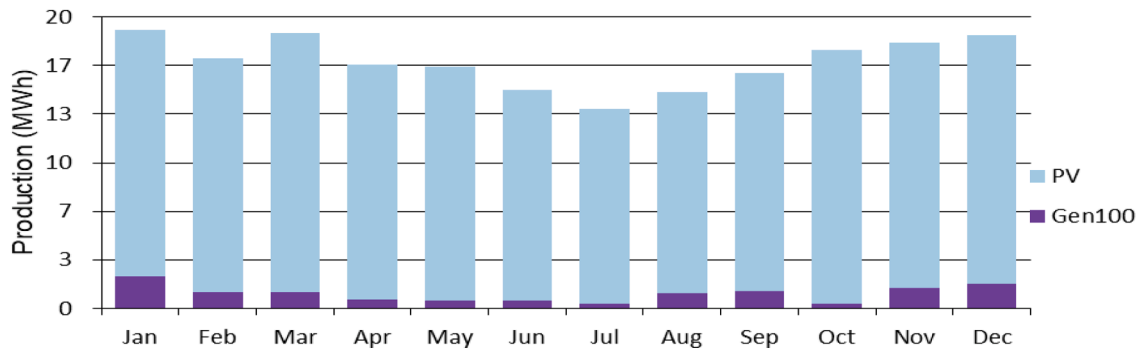


Fig. 8. Energy Generation of Proposed HES.

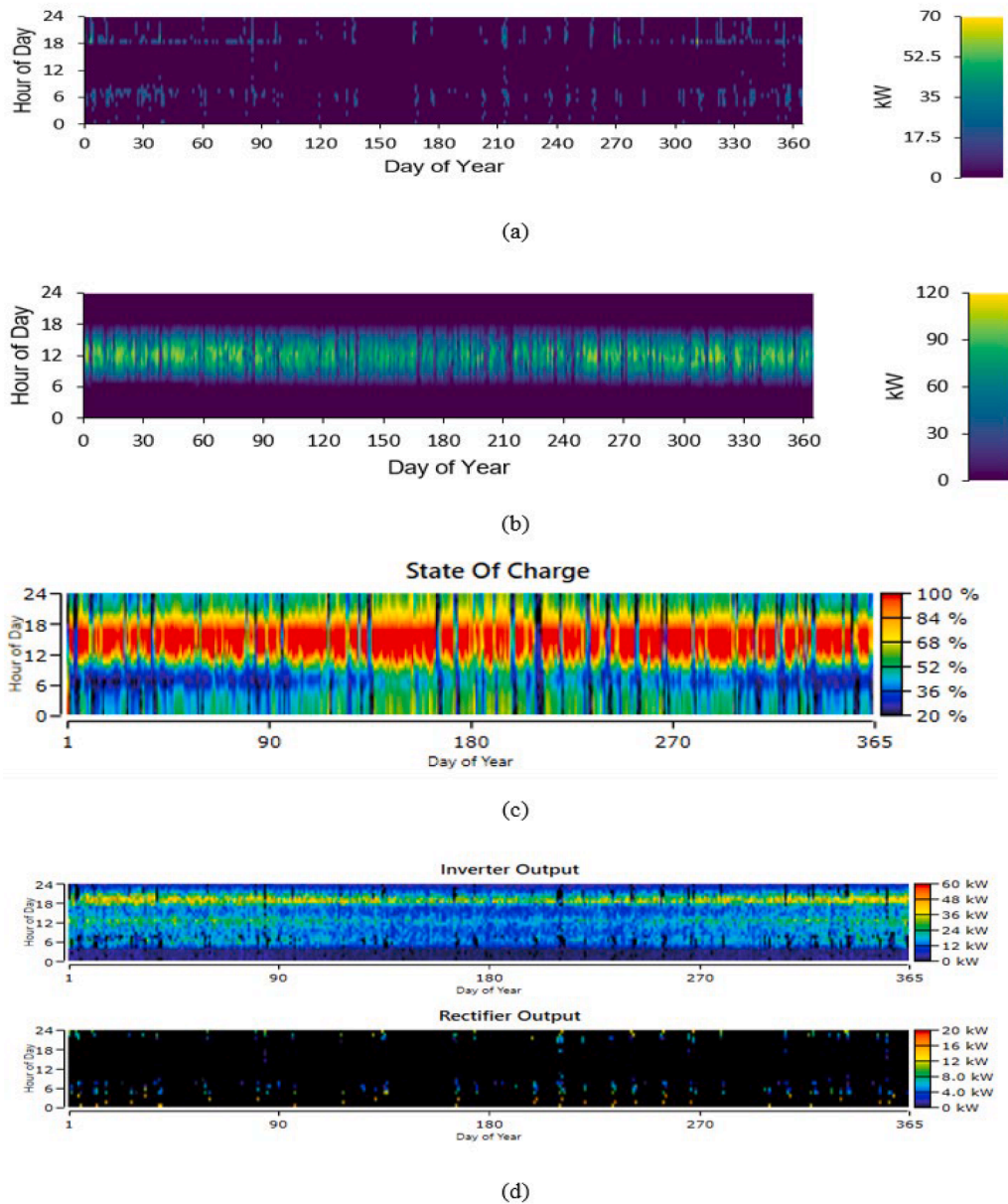


Fig. 9. AD out power for (a) DG (b) SPV (c) BS (d) Converter.

have also contributed to the contrasting LCOE results. The pandemic has disrupted global energy markets, leading to fluctuations in fuel prices. The surge in fuel prices, coupled with the inflation rate, has exerted

upward pressure on the LCOE of conventional energy sources, as observed in previous research. In contrast, the LCOE of solar PV systems has experienced a downward trend owing to declining PV module costs,

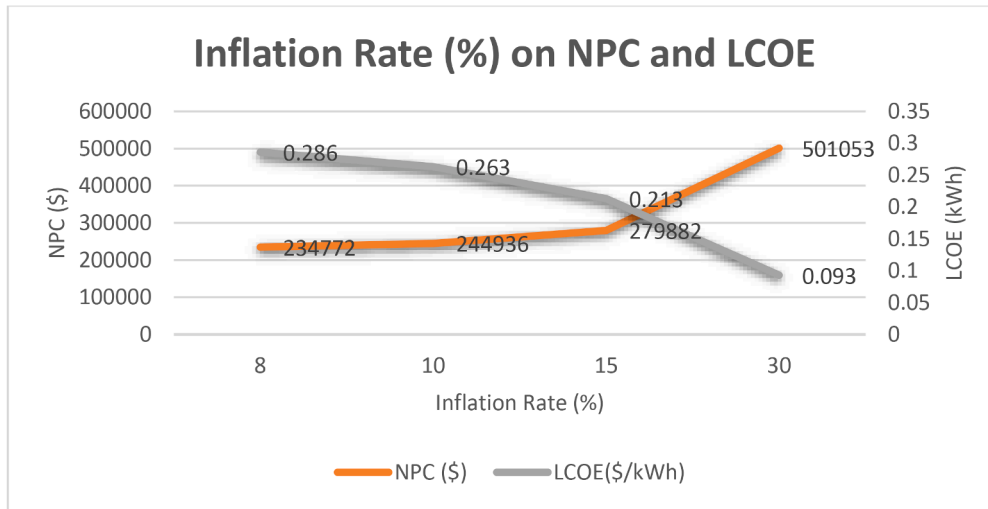


Fig. 10. Sensitivity analysis of inflation rate on NPC and LCOE.

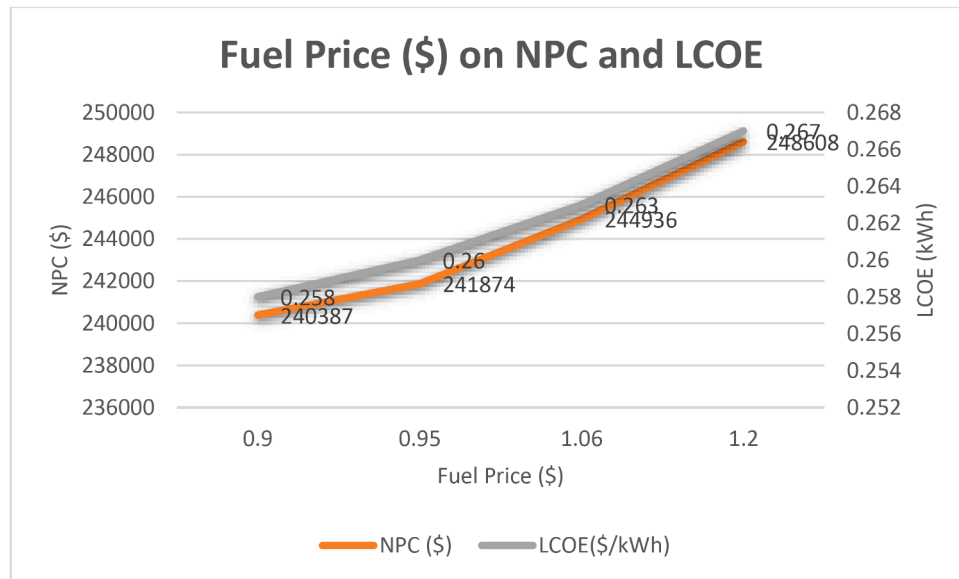


Fig. 11. Sensitivity analysis of fuel price on NPC and LCOE.

technological advancements, and increased economies of scale. Therefore, the recent changes in solar PV costs, along with the adverse effects of the COVID-19 pandemic on fuel prices and inflation rates, have influenced the outcomes of past studies and highlighted the importance of conducting up-to-date analyses. The present study considers these dynamic factors, providing a more accurate assessment of the optimal solution for rural electrification in Ghana, based on prevailing economic conditions and the evolving renewable energy landscape.

The integration of machine learning techniques marks a novel contribution, enhancing the accuracy of energy demand forecasting and ensuring the design of systems tailored to local energy needs. This approach not only improves system reliability but also ensures efficient utilization of resources, making it adaptable to diverse rural contexts. By incorporating advanced machine learning techniques, the study enhanced the accuracy of energy consumption predictions, which are crucial for the optimal design and operation of hybrid systems. The integration of advanced predictive analytics into traditional energy system design practices represents a critical step forward in the quest for universal access to reliable and clean energy.

Furthermore, the sensitivity analysis highlighted the resilience of the

proposed systems to fluctuations in key economic parameters. Variations in inflation rates, fuel prices, and discount rates influence the systems' financial outcomes, thereby demonstrating the need for adaptable strategies considering dynamic economic conditions. Finally, a comparison with grid-connected alternatives indicated that despite the apparent affordability of grid-connected electricity, the proposed standalone hybrid energy system offers a more reasonable solution for remote communities very far away from the grid. The Electric Distance Limit (EDL) exhibited that extending the grid to such areas is not financially viable, while standalone hybrid systems are the best alternative for rural electrification.

Beyond Ghana, the findings have broader implications for policy and practice in rural electrification globally. They underscore the importance of adopting hybrid systems to meet the dual challenges of energy access and sustainability. Policymakers and energy planners can leverage these insights to develop strategies that encourage the deployment of HES, supported by targeted incentives and investments.

Future studies should focus on employing advanced machine learning techniques, such as deep learning, to enhance energy demand forecasting and optimization. Long-term studies on environmental

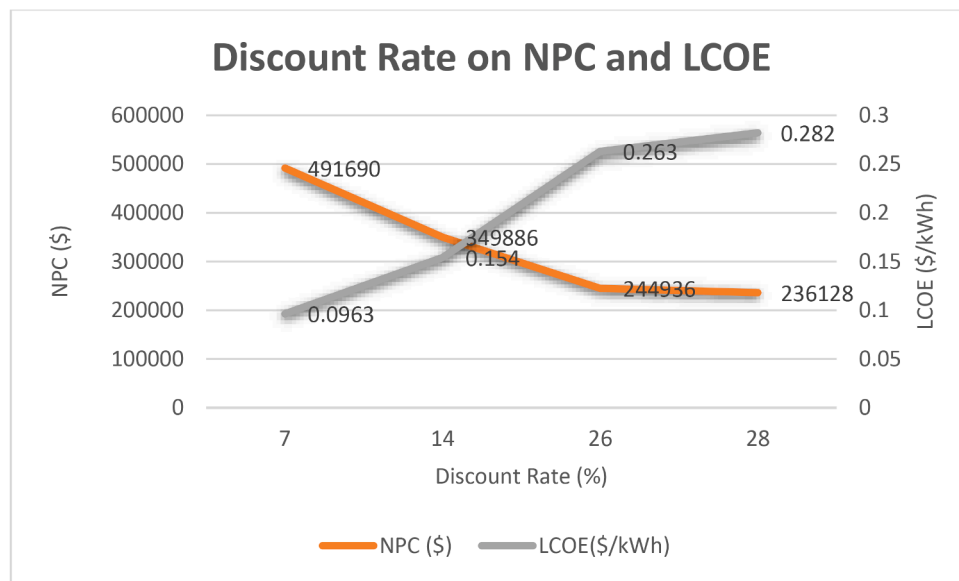


Fig. 12. Sensitivity analysis of discount rate on NPC and LCOE.

impacts, scalability, and replicability across different regions would provide deeper insights. Additionally, economic and policy analyses could identify strategies for encouraging investments and supporting HES adoption. Research on the resilience of hybrid systems to climate change would ensure their long-term sustainability. These directions build on this study's findings to advance the adoption of sustainable energy solutions and lay a strong foundation for improving energy access in off-grid and underserved regions.

4.1. Limitations of study

The study's primary limitations pertain to the constraints imposed by time and energy resource data sources, although the study objectives were achieved by the method utilized. First, owing to the limited timeframe of the study, the collection of localized energy resource data was impossible. This limitation led to reliance on data from sources such as NASA, which may not fully reflect the real conditions on the ground. Second, while the sensitivity analysis was performed on economic aspects, constraints once again hindered the examination of technical and environmental factors. These limitations should be acknowledged when interpreting the results and suggesting avenues for future research to explore these dimensions more comprehensively.

CRedit authorship contribution statement

Sufyan Yakubu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ravi Samikannu:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Investigation, Conceptualization. **Abid Yahya:** Supervision, Conceptualization. **Irfan Anjum Badruddin:** Visualization, Validation, Project administration, Funding acquisition. **Sarfaraz Kamangar:** Visualization, Validation, Project administration, Funding acquisition. **Maruliya Begam Kadarmydeen:** Conceptualization, Validation, Visualization. **Maxmilian Kwarteng:** Visualization, Validation, Supervision, Project administration, Conceptualization. **Nyagong Santino David Ladu:** Writing – review & editing, Validation. **Samuel Dodobatia Wetajega:** Validation, Resources, Data curation.

Declaration of competing interest

The authors declare that they have no competing interests.

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Data availability

Data will be made available on request. Data used to support the findings of this study are available from the corresponding author upon request.

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